SnapChef

TECHIN 515 | HWSW2

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https://github.com/IrisYzh/SnapChef

Problem Statement

Problem:

Keeping track of fridge ingredients is a small but persistent pain point for home cooks. Items get forgotten, groceries are bought twice, and the question "what can I make tonight?" often goes unanswered even when the fridge is full. SnapChef addresses this with on-device ingredient recognition and cloud receipt scanning. Together, they keep the fridge inventory up to date and suggest dishes the user can make based on what is currently available.

Intended User:

SnapChef is designed for home cooks who want a low-effort way to stay on top of what is in their fridge. This includes busy individuals and families who cook regularly but struggle to keep track of ingredients, as well as people trying to reduce food waste or cut down on unnecessary grocery trips.

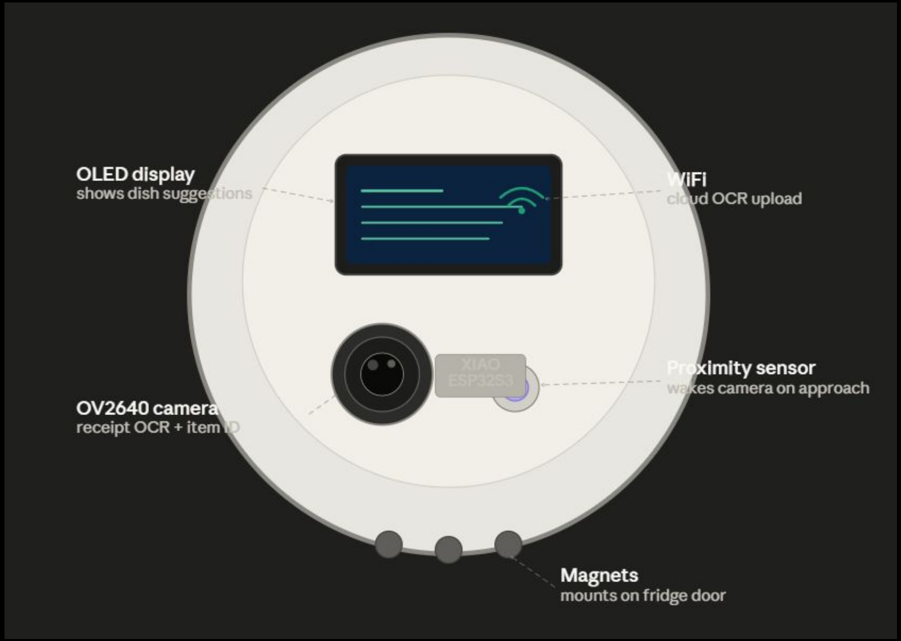
Solution Description

SnapChef is a circular, magnet-mounted smart device that attaches to any fridge door and automatically tracks the ingredients inside using a combination of on-device vision hardware and cloud OCR.

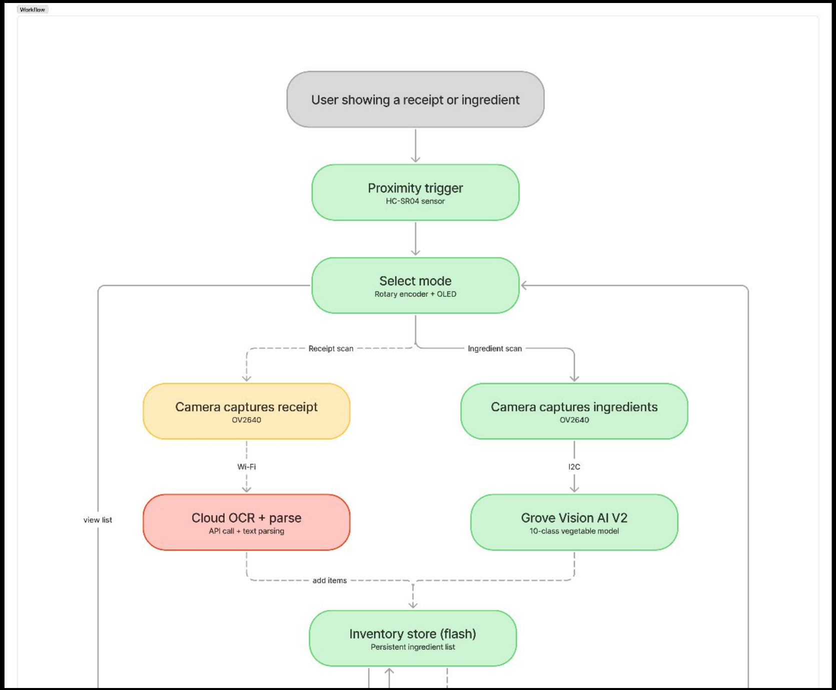
Solution

Key Features:

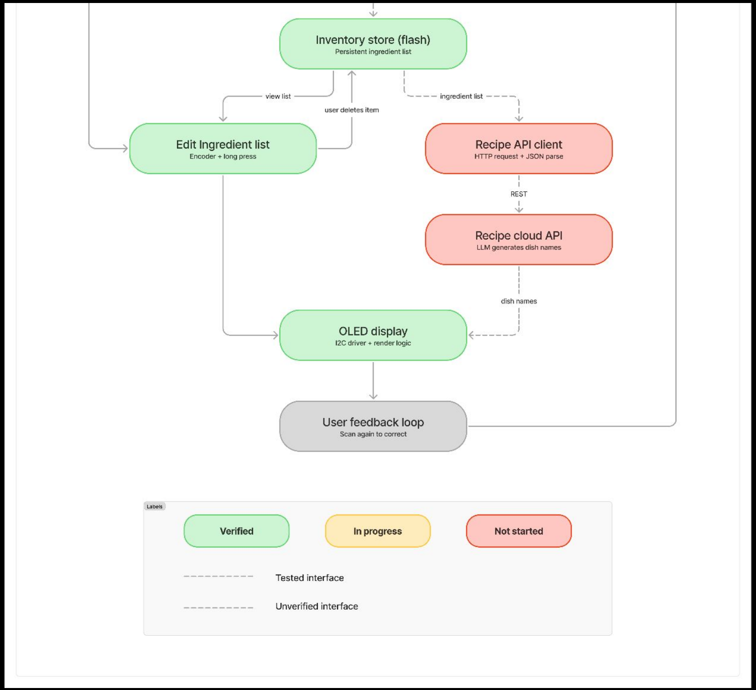
- **Receipt scanning.** The user holds a grocery receipt up to the camera, the device scan the image and automatically add the items to the inventory.
- **Ingredient recognition.** A proximity sensor wakes the camera when an ingredient is nearby. A Grove Vision AI V2 module identifies the item.
- **Dish suggestions.** After each scan, the OLED display shows a short list of dishes the user can make with what is currently in the fridge.
- **Manual editing.** The user can browse and remove items from the inventory at any time using the rotary encoder and buttons on the device.
- **Persistent storage.** The inventory is saved to flash memory and retained between sessions.



Update Architecture



Input



Output

OLED and Sensor Prototype

The prototype integrates HC-SR04, rotary encoder, mode button, and SSD1306 OLED on a breadboard with XIAO ESP32S3. All components were tested together and performed as expected.

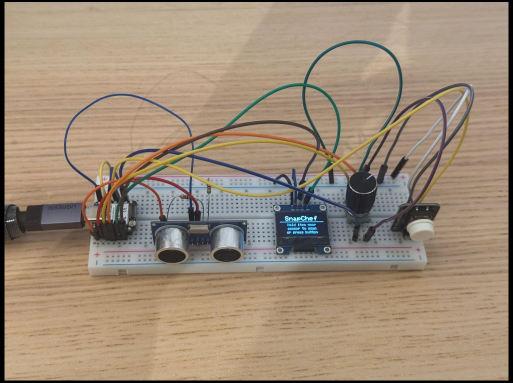
Component	Test	Result
HC-SR04	10 triggers at 10 cm	Good: 10/10 success
Rotary encoder	Scroll and mode select	Good: No missed inputs
State machine	10s / 30s timeout	Good: Triggered correctly in all cases
Long press delete	Hold 3s to remove item	Okay: Confirmed working
OLED	All 5 screens	Good: Rendered correctly

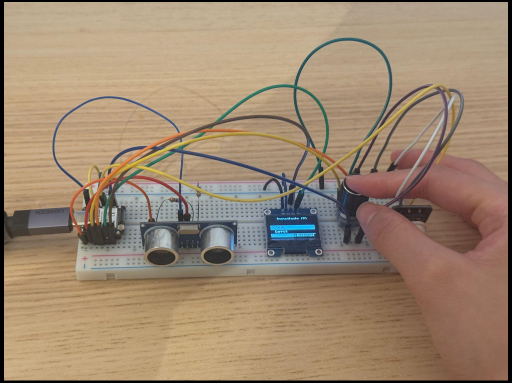
Known issue: the progress bar during long press causes the full screen to redraw each loop cycle, making the bottom hint text flicker. To be fixed by redrawing only the progress bar region.

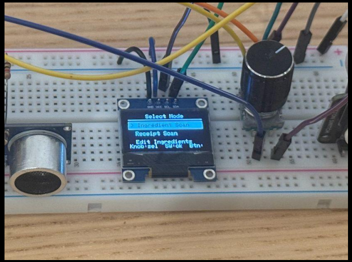
OLED and Sensor Prototype

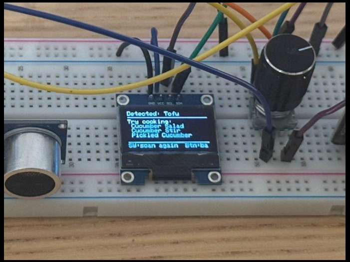
Future Works:

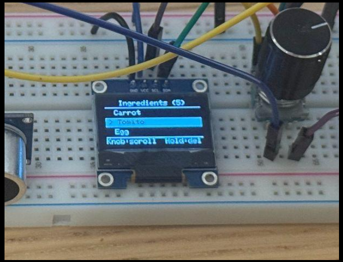
1. Fix the long-press progress bar flicker by redrawing only the progress bar region instead of refreshing the full screen.
2. Connect the dish suggestion screen to a recipe API that takes the current inventory as input and returns 2-3 personalised dish recommendations based on what the user actually has.
3. Work on receipt recognition







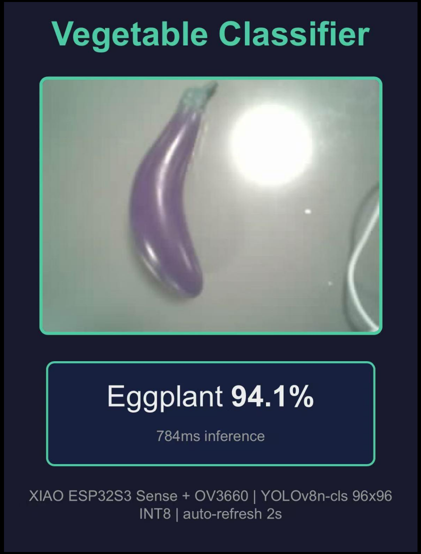


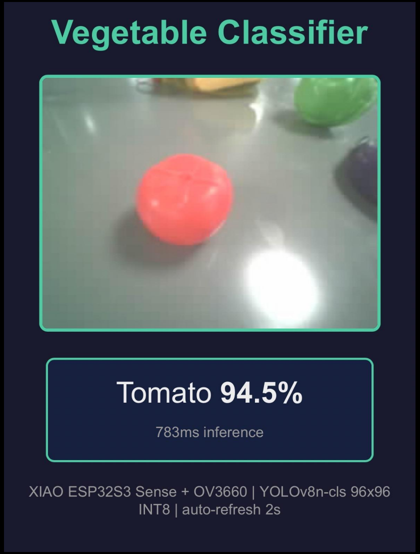


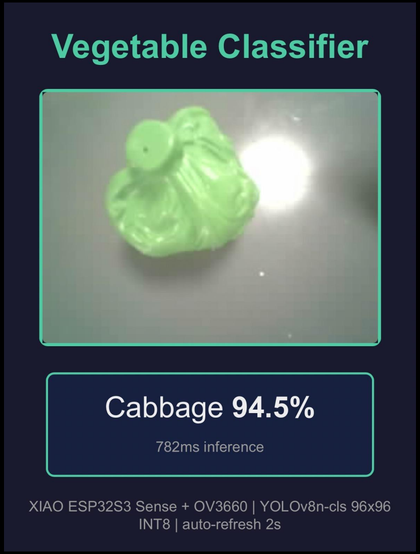
Ingredient Classification

10 Classes: Bell Pepper, Broccoli, Cabbage, Carrot, Cucumber, Eggplant, Garlic, Onion, Potato, Tomato
Dataset: Training samples 877, Validation samples 105
Input Size: 96*96*3

Trade-off: **Class Number** vs **Accuracy** vs **Inference Time**







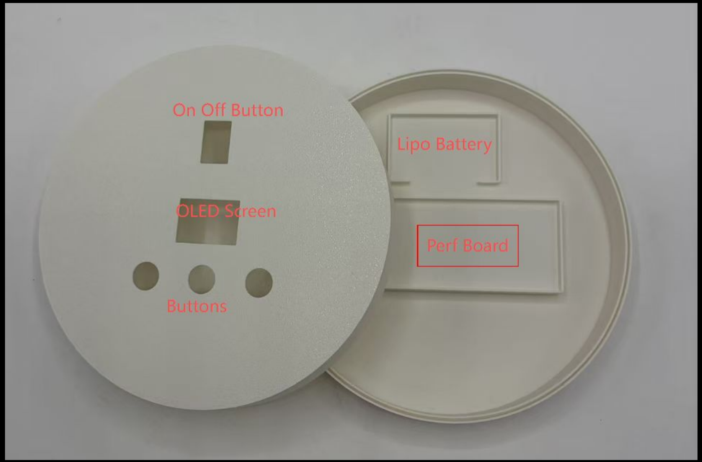
Ingredient Classification

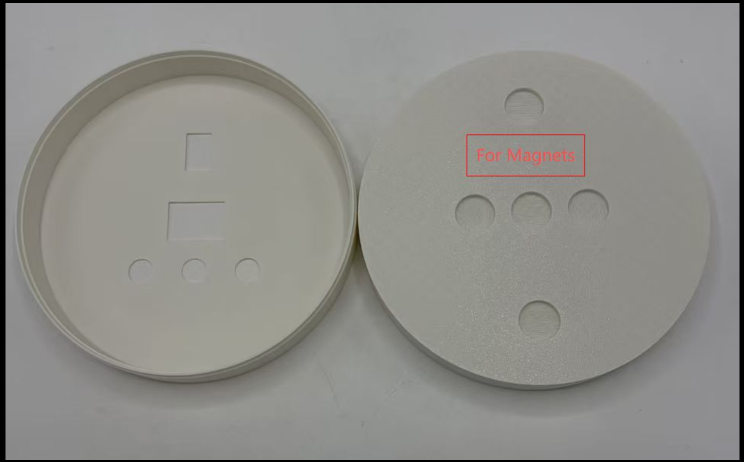
Device	Model	Accuracy	Accuracy (Int 8)	Inference Time	Model Size
ESP32S3 Sense	CNN	78.10%	74.29%	33ms	23.1 KB
ESP32S3 Sense	Yolo V8	97.14%	93.33%	782ms	1.45 MB
ESP32S3 Sense	MobileNet V2	77.11%	62.05%	248ms	624 KB
Grove AI Vision V2	SenseCraft	-	-	17ms	291 KB

Future Work:

1. Fix the I2C communication problem between Grove and ESP32, move the model to Grove
2. Add more classes, including meat, eggs...
3. Collect more data and improve model accuracy

Enclosure Prototype





Future Works:

1. Optimize the component layout based on the chosen hardware.
2. Assemble the perfboard to replace the breadboard prototype.
3. Explore larger display options such as a touchscreen or e-ink display to improve readability and potentially simplify the input controls.

Budget Update

Item Name	Quantity	Unit Price	Total Price	Supplier	Link
Seeed Studio XI	2	$23.51	$47.02	Amazon	Amazon.com: Se
20 Pieces Play V	1	$19.79	$19.79	Amazon	Amazon.com: 20
Grove - Vision A	1	$24.49	$24.49	Amazon	https://www.ama
Gikfun 12x12x7.	1	$8.53	$8.53	Amazon	https://www.ama
Twidec/5Pcs Ro	1	$5.99	$5.99	Amazon	https://www.ama
Strong Ceramic	1	$9.99	$9.99	Amazon	https://www.ama
Total	7		$115.81		